Thicker but Narrower? Weakly Supervised Bounds on Opportunity Exposure in Urban Ride-Pooling Markets

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ABSTRACT

Ride-pooling algorithms jointly determine operational efficiency and which travelers could share a service chain, but public mobility releases suppress the group identifiers needed to study that exposure margin. We introduce BoundPool, a weakly supervised structured-inference framework for bounding neighborhood-level *opportunity exposure* from privacy-coarsened trip records. Trips are nodes; temporal, origin–destination, and directional constraints define candidate pairs. A multiple-instance noisy-OR model learns edge hazards using only trip-level match indicators. We then optimize socioeconomic mixing statistics over all exact-cover packings consistent with observed match status, reporting an admissible interval rather than a single imputed co-rider graph. In a locked Chicago-coordinate synthetic benchmark, a geography-equality-free score improves held-out node Brier loss by 90.2% over a transparent rule, raises hidden-edge mean reciprocal rank from 0.781 to 0.941, and retains all 80 hidden pairs. Its 90% score-retention interval is 34.5% narrower than the untrimmed range and covers truth; its 95% interval is 58.6% narrower and also covers. A diagnostic map containing tract-equality indicators appears sharper but mechanically encodes the synthetic same-income target and excludes truth at 95%, exposing proxy-induced false precision. We pair the method with a reproducible design for Chicago's 2026 ride-pooling policy change and publicly released TNP and ACS data. The current draft reports the completed method validation and pre-policy feasibility audit; complete-day Chicago structural and causal estimates are explicitly reserved for the next data run. The framework studies compatibility-weighted opportunity exposure, not observed co-presence, individual socioeconomic status, or echo chambers.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Geographic information systems*; • **Applied computing** → *Transportation.*

KEYWORDS

ride pooling, weak supervision, partial identification, set packing, urban mobility, socioeconomic exposure, privacy-coarsened data

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1 INTRODUCTION

Urban pooling platforms use prediction and combinatorial optimization to decide which requests can share a vehicle. These decisions do more than reduce vehicle miles: they define a technologically mediated set of potential cross-neighborhood encounters. The scientific question is therefore not only whether a thicker pooling market produces more matches, but whether the additional feasible matches broaden or narrow socioeconomic opportunity exposure.

That question is difficult precisely where public evidence is most available. The City of Chicago releases trip-level Transportation Network Provider (TNP) records with rounded timestamps, coarse origins and destinations, sharing authorization, realized match status, and the number of component trips in a pooled service chain. It does not release the chain identifier, vehicle identifier, provider, or stable rider identity. Treating nearby trips as known co-riders would convert privacy coarsening into false certainty. Conversely, discarding the relational object leaves only an ordinary match-rate regression and cannot measure the exposure margin.

We propose a third route: infer an *admissible set* of latent groupings and carry that ambiguity into the scientific estimand. BoundPoolfirst builds a sparse candidate graph from physically interpretable constraints. It then learns edge compatibility from node-level match labels through a multiple-instance likelihood, without constructing pair labels. Finally, a set-packing program finds the minimum and maximum socioeconomic mixing statistics over feasible groupings. The result is conditional on declared candidate and score restrictions, but it makes those assumptions visible and testable.

Our contributions are fourfold. First, we formulate suppressed pooling groups as a structured missing-data problem rather than a clustering target. Second, we develop a scalable weak-supervision objective that converts public node outcomes into edge hazards while preserving the absence of edge truth. Third, we replace a single reconstructed network with optimization-based opportunity-exposure bounds and diagnose the coverage cost of aggressive model restriction. Fourth, we provide a human-experiment-free empirical design that combines Chicago TNP records, tract-level ACS covariates, and the January 2026 pooling-policy change.

The locked pilot supplies an important negative result as well as a positive one. A full feature map sharply improves node calibration but uses tract equality that becomes an algebraic proxy for the synthetic same-income target; its 95% score floor misses truth. Removing six geography-equality features preserves node performance, improves hidden-edge ranking, and restores coverage, although continuous coordinates can still proxy SES. AI therefore organizes model-indexed admissible sets, while the score-free range remains primary.

2 RELATED WORK AND POSITIONING

Shareability and dynamic pooling. The algorithmic ride-pooling literature usually begins from requests whose origins, destinations, and time constraints are observed. Santi et al. construct a *shareability network*: two trips are adjacent when one vehicle can serve both within a delay budget, and a maximum matching translates this graph into an upper bound on fleet and trip reductions [12]. Alonso-Mora et al. extend this logic to online, capacitated service by enumerating feasible request groups and repeatedly solving a trip–vehicle assignment problem [3]. Complementarily, Ozkan and Ward study forward-looking dynamic matching and establish large-market optimality guarantees for policies that account for spatially heterogeneous arrivals and patience [10]. These papers solve a forward allocation problem on an observed request system. Our scientific problem is inverse: the public Chicago file retains node-level authorization and match outcomes but suppresses the co-rider group identifier. We therefore infer a sparse set of candidate edges from coarsened time and origin–destination constraints, learn their compatibility from node labels, and retain all set packings consistent with the public record. The target is not a single operational assignment but bounds on a latent, privacy-removed urban mixing quantity. This distinction matters because timestamp rounding can make many pairwise links individually plausible while rendering them jointly impossible. Graph feasibility and global matching constraints are therefore part of the estimand, not merely computational devices for accelerating a conventional matcher.

Efficiency, fairness, and the Chicago evidence. Algorithmic fairness work has modified ride-pooling objectives to balance service and distributional goals. Raman, Shah, and Dickerson incorporate rider- and driver-side fairness into pooling allocation and study driver-income redistribution using New York taxi data [11]. This is an important design perspective, but it treats fairness as a criterion imposed on a known matching problem. We instead measure the socioeconomic composition of the *opportunities* created by pooling when the realized matching graph is not released. Thus, our estimand is neither a driver-income fairness objective nor a claim that two inferred riders actually interacted.

Several papers use the same Chicago administrative source and are consequently the closest empirical benchmarks. Taiebat, Amini, and Xu predict whether riders request sharing and whether authorized trips are successfully matched, finding that travel impedance variables dominate their trip-level machine-learning models [14]. Mucci and Erhardt use travel-time, cost, and choice models to distinguish the decision to request a shared ride from the subsequent match outcome [8]; their published corrigendum cautions against using the original cross-elasticity calculations. Most directly, Sebti and Chen cluster authorized Chicago trips under spatiotemporal rules and train logistic, random-forest, and XGBoost classifiers for the authorization-to-match gap [13]. Those studies establish that node-level sharing and matching are predictable. They do not identify co-rider edges, enforce global set-packing consistency, or propagate edge ambiguity into socioeconomic exposure bounds. A node classifier alone would therefore not be a contribution of our paper; the contribution is structured inference under a known privacy observation operator.

Chicago's differentiated transportation-network-company tax also has an established causal literature. Abkarian, Hegde, and Mahmassani use interrupted time series and a Bayesian hierarchical model and report that the January 2020 tax differential shifted demand toward shared trips, although against a declining baseline [2]. Zheng et al. use difference-in-differences and find lower overall TNC ridership, growth in shared trips, decline in solo trips, and substantial spatial heterogeneity [17]. These studies identify changes in trip counts or shares. Our proposed policy analysis asks a different downstream question: conditional on an intervention thickening the authorized pool, did the set of feasible cross-neighborhood encounters broaden or become more socioeconomically assortative? Because the latent groups are not observed, that question must be answered with lower and upper bounds rather than with an imputed network treated as truth.

Urban mobility and experienced segregation. This estimand is motivated by a literature showing that residential segregation and experienced segregation are not interchangeable. Wang et al. use geolocated mobility to show that residents of Black and Hispanic neighborhoods in large U.S. cities visit a systematically different composition of neighborhoods even when travel radius and spread are similar [16]. Moro et al. combine mobile-device visits, venues, and home-area income proxies to measure place and individual experienced income segregation; mobility and social exploration help explain large within-city variation [7]. Nilforoshan et al. move from shared places to a fine-grained person-to-person exposure network and show that large metropolitan areas can be more exposure-segregated, while hubs that bridge diverse neighborhoods are associated with lower segregation [9]. Abbiasov et al. identify a related efficiency–mixing tension: more local "15-minute" consumption is associated with higher experienced segregation for low-income residents [1]. Most recently, Fan et al. use official travel surveys in five cities, including Chicago, and show that replacing self-reported income with home-neighborhood income yields systematically lower estimated mixing [6].

These findings supply the substantive hypothesis but also delimit what our data can support. Unlike mobile-phone panels or travel diaries, Chicago's public TNC records contain neither persistent persons nor verified co-presence, and ACS median household income is a neighborhood attribute rather than rider income. We therefore use the terms *compatibility assortativity*, *opportunity exposure*, and *mobility silo*; we do not claim realized social contact, durable homophily, or an echo chamber. Methodologically, our bridge between the two literatures is a privacy-coarsened latent graph: weak supervision maps node-level match labels to edge scores, exact-cover constraints enforce feasible pair structure, and optimization carries the remaining structural uncertainty into sharp or sensitivity-indexed exposure bounds. This makes AI scientifically necessary only insofar as it tightens calibrated bounds relative to transparent compatibility rules while preserving coverage on known-truth simulations.

3 DATA AND SCIENTIFIC SETTING

3.1 Chicago TNP records

The City of Chicago's 2025– TNP release contains one record per customer trip [4] and includes start and end timestamps rounded to 15 minutes, trip distance and duration, pickup and drop-off census tracts or community areas, centroid coordinates, fares and charges,

Table 1: Completed audit of the Chicago prefix extract.

Quantity	Measured value
Trips	53,241
Shared option authorized	3,048 (5.72%)
Matched / all trips	2,146 (4.03%)
Matched / authorized	70.41%
Matched without authorization	0
Matched with `trips_pooled`< 2	83
Missing pickup / drop-off tract	38.05% / 38.39%
Missing pickup / drop-off centroid	8.98% / 9.21%
ACS join given released tract (PU / DO)	95.24% / 94.99%

Note. Measured in the frozen 53,241-row identifier-prefix extract; not population estimates.

plus authorization, match, and pooled-chain-size fields. The last field, `trips_pooled`, counts all customer trips from the time the vehicle was empty until it became empty again; two component riders need not have been simultaneously present. Each component has a separate row and the same count, but the common chain ID is suppressed. Accordingly, our pair estimand is a service-chain compatibility opportunity, not physical co-presence.

A deterministic two-character trip-ID-prefix sample covers December 9, 2025 through February 2, 2026 and contains 53,241 trips, including 3,048 authorized and 2,146 matched records. It is suitable for schema tests and aggregate-rate feasibility, but not latent-edge reconstruction: conditional on one component being sampled, its counterpart is omitted with probability approximately 255/256 under the sampling device. Structural estimation therefore requires complete narrow day slices. The completed audit finds 70.41% matched among authorized trips, zero matches without authorization, and 83 matched records with `trips_pooled`< 2. Pickup tracts and centroids are missing for 38.05% and 8.98% of rows; the corresponding drop-off rates are 38.39% and 9.21%. Conditional on release, 95.24% of pickup and 94.99% of drop-off tracts join to the ACS file. These quantities describe this prefix extract, not the Chicago population.

3.2 Socioeconomic proxies

We join pickup census tracts to the 2024 five-year American Community Survey Detailed Tables B19013 and B03002 [15]. The prepared extract contains 1,551 Chicago-area tracts and supplies median household income and race/ethnicity composition. Income and composition refer to the trip's origin neighborhood, not to the rider. The primary pair statistic uses ACS income quintiles; continuous sensitivity analyses use absolute log income differences. Rows without a released and joined pickup tract remain in match-outcome estimation but not in the primary SES bound.

3.3 Policy setting and completed feasibility signal

Chicago changed its congestion-zone surcharge structure on January 6, 2026 [5]. Inside the new zone, the solo–shared surcharge gap is $0.90 during weekday active hours and $1.50 on weekends. Our planned estimand is the policy intention-to-treat effect on authorization, successful pooling, and the opportunity-exposure

Table 2: Raw policy-window feasibility signals (not causal).

Cell	Period	*N*	Authorized	Matched/all	Add'l charges
Target	Pre	3,797	5.95%	4.61%	$5.69
Target	Post	4,047	8.08%	6.89%	$6.31
Comparison	Pre	10,975	5.53%	3.54%	$6.86
Comparison	Post	11,369	5.82%	3.97%	$6.85

Note. Four-week windows around January 6, 2026; approximate feasibility-stage zone; no sampling uncertainty or causal adjustment.

Table 3: Data layers and current completion status.

Layer	Status	Scientific use
Chicago prefix sample	Complete	Schema, missingness, aggregate feasibility only
Chicago complete days	**TBD**	Candidate graph, held-out node calibration, SES bounds
ACS 2024 tracts	Complete	Pickup-neighborhood SES proxy
NYC HVFHV	Pilot extract	Time-coarsening and out-of-domain stress test; no pair truth
Locked synthetic market	Complete	Candidate recall, hidden-edge ranking, coverage audit

Note. Red TBD entries require complete-day or full-month data and must be replaced by measured results before submission.

interval, using a post-by-zone-by-active-hours design. The policy changes both market thickness and selection into the shared pool, so thickness is a mechanism, not an excluded instrument.

In the prefix feasibility sample, the approximate newly covered zone during weekday active hours shows authorization increasing from 5.95% to 8.08% and matched/all trips from 4.61% to 6.89%. Mean additional charges increase from $5.69 to $6.31 in the treated cell. These are unadjusted go/no-go signals, not causal estimates. Table 3 separates completed facts from the complete-day analysis still required.

3.4 Frozen data-management protocol

The production downloader issues a server-side count for each complete-day query, retrieves every authorized row in deterministic identifier order, paginates until that count is met, writes through an atomic temporary file, and records query text, retrieval time, row count, and SHA-256 digest. A schema audit then checks identifier uniqueness, exact date boundaries, boolean-field consistency, pooled-chain codes, coordinate ranges, and tract suppression. We do not combine a partial download with a complete day. The raw response, processed node table, and machine-readable audit ledger remain separately versioned.

Days, not edges, are the train/test unit. Candidate construction is performed independently inside each date and split, preventing an edge or rounded-time duplicate from leaking across evaluation sets. ACS joins occur only after the physical candidate graph is frozen, and SES never enters candidate support or weak-MIL fitting. The primary SES analysis is complete-case at pickup tract; prespecified alternatives use community-area aggregates and an explicit suppressed-location stratum. Every model result is paired with the eligible denominator, candidate-supported denominator, degree distribution, exact-cover infeasibility rate, and candidate-threshold configuration.

3.5 NYC temporal-coarsening stress test

New York City TLC high-volume for-hire-vehicle records provide a distinct measurement environment with second-level request, pickup, and drop-off times but only coarse taxi zones. A completed schema pilot contains 2,000 deliberately stratified records: 500 legacy rows from 2019–2020 and 1,500 modern rows from February 2021 and January 2023. Every modern pilot row has non-missing request, pickup, drop-off, and zone fields. The audit flags 0.8% negative waits in one 2021 stratum and four of 250 matched 2023 records without a shared-request flag. Because the extract is outcome-stratified, it cannot estimate match rates. The planned full-month test will round timestamps to 5, 15, and 30 minutes and measure graph inflation and node calibration. It cannot validate true pairs because NYC also suppresses rider, vehicle, and shared-group identifiers.

4 WEAKLY SUPERVISED STRUCTURED INFERENCE

4.1 Observed nodes and latent groups

Let $V = \{1, \ldots, n\}$ index authorized trips in a complete market-day slice. Trip i has released features x_i, node match label $y_i \in \{0, 1\}$, released service-chain size k_i, pickup-neighborhood log income Z_i, and predeclared income bin B_i. The latent object is a set M of mutually non-conflicting pairs. For the primary pair analysis, we retain $V_1 = \{i : y_i = 1, k_i = 2\}$ with valid geography and require every retained node to appear exactly once. If $m = |V_1|/2$, the target statistics are

$$H(M) = m^{-1} \sum_{\{i,j\} \in M} \mathbb{I}\{B_i = B_j\}, \quad D(M) = m^{-1} \sum_{\{i,j\} \in M} |Z_i - Z_j|. \tag{1}$$

H is a same-neighborhood-income-bin share, not degree-corrected network assortativity; D is a continuous sensitivity outcome. Neither is observed because M is suppressed.

4.2 Candidate construction

We create an undirected candidate edge (i, j) only within a calendar day and train/test split when start-time gap is at most 30 minutes, pickup and drop-off distances are at most 4 and 7 km, and OD-vector cosine is at least -0.25. A normalized spatiotemporal nearest-neighbor query proposes at most 96 records; physical cost ranks proposals, and a deterministic cap limits degree to 16. The resulting candidate set is E_C. The production-primary 22-column feature map ϕ_{ij} contains normalized time, OD, duration, mileage, direction, interval overlap, squares, exponentials, and predeclared interactions. A 28-column diagnostic map adds six same-tract/same-area indicators and interactions. Neither endpoint match status, `trips_pooled`, fare, nor ACS SES enters either map.

Search and degree caps are computational restrictions that can delete the truth. We therefore report support, degree, and threshold sensitivity; true candidate recall is knowable only in synthetic validation.

4.3 Node-label weak supervision

The edge model assigns logit $a_{ij} = \phi_{ij}^\top \theta$ and activation score $q_{ij} = \sigma(a_{ij})$. This score is not a calibrated posterior that the endpoints shared a vehicle. No edge label is available. Instead, incident edges form a bag for node i and induce a noisy-OR node probability

$$\begin{aligned} \widehat{p}_i(\theta) &= 1 - \prod_{j:(i,j) \in E_C} (1 - q_{ij}) \\ &= 1 - \exp\left\{ - \sum_{j:(i,j) \in E_C} \operatorname{softplus}(a_{ij}) \right\}. \end{aligned} \tag{2}$$

On candidate-supported training nodes $\mathcal{T}$, we estimate

$$\widehat{\theta} \in \arg\min_{\theta} \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \{-y_i \log \widehat{p}_i - (1 - y_i) \log(1 - \widehat{p}_i)\} + \frac{\lambda}{2} \|\theta_{-0}\|_2^2, \tag{3}$$

with $\lambda = 10^{-3}$, unpenalized intercept, stable softplus evaluation, and L-BFGS-B optimization. The current implementation uses a linear model over a fixed nonlinear basis of time, OD, directional, overlap, and interaction features. A transparent rule combines the same physical dissimilarities with disclosed fixed weights; only its intercept and a nonnegative global scale are calibrated through Eq. (3). Thus the comparator tests flexibility beyond an interpretable physical ordering, not performance against an edge-supervised oracle. Whole days, rather than edges, form the train/test split.

Equation (2) supports a node prediction claim only. Two matched endpoints may belong to different service chains, so public-data pair accuracy is undefined. Synthetic truth is used exclusively after fitting to audit candidate recall and within-node edge ranks.

4.4 Set-packing opportunity-exposure bounds

For candidate edge e, let $z_e \in \{0, 1\}$ indicate selection, $w_e \geq 0$ be its compatibility score, and g_e be an SES statistic such as $\mathbb{I}\{B_i = B_j\}$ or $|Z_i - Z_j|$. With required matched nodes V_1, the feasible pair packings are

$$\begin{aligned} \mathcal{M}(E_C) = \{z \in \{0, 1\}^{|E_C|} : \sum_{e \ni i} z_e &= 1 \quad \forall i \in V_1, \\ z_e &= 0 \quad \text{if } e \not\subseteq V_1\}. \end{aligned} \tag{4}$$

Because every feasible solution selects $|V_1|/2$ edges, the untrimmed admissible interval is

$$\left[\frac{2}{|V_1|} \min_{z \in \mathcal{M}(E_C)} \sum_e g_e z_e, \frac{2}{|V_1|} \max_{z \in \mathcal{M}(E_C)} \sum_e g_e z_e \right]. \tag{5}$$

For sensitivity, let $W^* = \max_{z \in \mathcal{M}(E_C)} \sum_e w_e z_e$. A ρ-retention set adds $\sum_e w_e z_e \geq \rho W^*$. This can tighten the interval but is model-based and may lose coverage; it is not a confidence interval. All programs are solved as binary linear programs with HiGHS and reported only after a global optimality certificate.

Proposition 1 (Conditional sharpness and coverage). *If $\mathcal{M}(E_C)$ is nonempty and both programs in Eq. (5) are solved globally, their endpoints are attained and sharp over the declared candidate graph. If the suppressed true pairing $z^\dagger$ lies in $\mathcal{M}(E_C)$, the untrimmed interval contains its target value. A ρ-restricted interval has the same guarantee only if $\sum_e w_e z_e^\dagger \geq \rho W^*$.*

Table 4: Locked synthetic holdout: node and edge diagnostics.

Metric	Rule	Primary AI	Diagnostic AI
Node Brier loss	0.06048	**0.00595**	0.00513
Node log loss	0.25443	**0.04732**	0.04206
Node ROC–AUC	0.9603	**1.0000**	1.0000
True-edge MRR	0.781	**0.941**	0.775
True edge top-1	67.5%	**90.0%**	64.4%
True edge top-3	85.6%	**98.1%**	90.0%
Candidate recall		80/80 (100%)	

Note. Ranking conditions on the hidden true edge being in the candidate graph; no edge labels enter training. Primary AI removes six geography-equality columns; diagnostic AI retains all 28 columns.

The proof is immediate from the one-to-one correspondence between feasible binary incidence vectors and exact-cover pairings; it is given with the full pseudocode in the appendix. Because larger ρ values form nested subsets, their widths weakly decrease, but Proposition 1 shows why narrowing alone gives no coverage guarantee.

5 VALIDATION RESULTS

5.1 Locked end-to-end benchmark

Before the first held-out run, we fixed a two-day Chicago-coordinate generator with 80 hidden pairs and 80 unmatched authorized trips per day, 15-minute time rounding, model defaults, and score retentions $\rho \in \{0.90, 0.95\}$. Pair IDs and synthetic income bins were written to separate truth files and never read by model fitting. The first day trains the model and the second is evaluated once.

The candidate graph retains 80/80 hidden holdout pairs. On all 240 supported holdout nodes, the transparent rule obtains Brier loss 0.06048 and the 22-feature primary weak-MIL score 0.00595, a 90.2% improvement. Its hidden true-edge MRR is 0.941 and top-1 rate 90.0%, versus 0.781 and 67.5% for the rule. By contrast, the 28-feature diagnostic score is slightly better at its trained node task but much worse at edge ranking because tract equality is a shortcut in this generator.

The hidden same-income-bin share is 0.5625 and the untrimmed interval is $[0.050, 0.775]$. The primary score yields $[0.288, 0.763]$ at $\rho = 0.90$ and $[0.388, 0.688]$ at $\rho = 0.95$, reducing width by 34.5% and 58.6%; both retain the complete hidden packing and cover truth. The diagnostic map appears much sharper, but its 95% region $[0.613, 0.775]$ excludes both. The contrast in Figure 1 reveals feature-induced rather than scientifically earned precision.

5.2 Coarsening and solver coverage

Across 20 independent small-market replications, candidate recall and raw-bound coverage are 100% for released time bins of 1, 5, 15, and 30 minutes. The candidate-edge/true-edge ratio rises from 1.04 to 2.11 and the mean raw bound width from 0.003 to 0.150 as timestamps are coarsened. Under the deliberately well-specified 95% rule-score restriction, mean width at 15 minutes falls from 0.073 to 0.020 and at 30 minutes from 0.150 to 0.027, with 100% coverage in this implementation check. These results validate the optimization logic, not the realism of the Chicago candidate model.

5.3 Prefix-only mechanics check

For completeness, we ran the pipeline on the structurally incomplete Chicago prefix graph. It creates only 151 candidate edges and supports 246 of 3,048 authorized nodes (8.1%). On the last-14-day holdout, just 93 supported nodes remain: the transparent rule has Brier score 0.15525 and weak MIL 0.15777, so the AI is 1.6% worse. This is a useful negative software check, not an estimate of Chicago performance. Under approximately independent prefix retention, the other component of a two-trip chain is absent with probability 255/256; poor support is therefore generated by the observation design rather than evidence that physical matching constraints fail. We do not run exact-cover exposure bounds on this graph or tune the candidate rule to improve these numbers.

5.4 Required Chicago results

The empirical paper will not replace the following cells with prefix-sample estimates. They require complete authorized-trip days and a pre-declared candidate-sensitivity grid.

6 ROBUSTNESS AND FAILURE PROTOCOL

Candidate support. Real data cannot reveal candidate recall, so support diagnostics must carry more weight than a conventional hyperparameter sweep. For every date and treatment cell we report authorized-node support, matched-node support, candidate degree quantiles, component sizes, and the share of pair-eligible components admitting an exact cover. The same summaries are stratified by pickup-income bin and tract-suppression status. Large support shifts at the policy boundary would contaminate the exposure contrast even if node Brier loss improved. We therefore run the candidate grid declared above without using H to choose a preferred graph. The untrimmed range remains the primary structural result because it relies on physical admissibility but not on a learned score.

Geography-proxy ablation. The locked 28-column score includes same-tract and same-area indicators. An edge audit shows that pickup- and drop-off-tract equality coincide exactly with the same-income-bin target on all 560 held-out matched-node candidates; area equality is constant. Removing the six equality columns leaves the 2,640 edge candidate graph unchanged. The resulting 22-feature model retains a 90.2% node-Brier gain, raises true-edge MRR from 0.781 for the rule to 0.941, and covers truth at both score floors while narrowing the raw interval by 34.5% and 58.6%. The full map's extra sharpness is therefore circular in this generator and is diagnostic only.

This completed ablation does not eliminate every proxy: synthetic income bins also generate fixed coordinate offsets, so continuous OD distances carry SES information. The production analysis additionally fits a continuous-geometry-only score and permutes ACS attributes across tracts after model fitting. If the AI width advantage or policy sign exists only under geographic proxies, we report geographic sharpening rather than a substantive AI contribution.

Table 5: Locked holdout: same-income-bin sensitivity regions.

Admissible set	Interval	Width	Reduction	Contains value	True packing eligible
Untrimmed candidate graph	[0.050, 0.775]	0.725	—	Yes	Yes
Rule, $\rho = 0.90$	[0.338, 0.775]	0.438	39.7%	Yes	Yes
Primary AI, $\rho = 0.90$	[0.288, 0.763]	0.475	34.5%	Yes	Yes
Rule, $\rho = 0.95$	[0.438, 0.750]	0.312	56.9%	Yes	No
Primary AI, $\rho = 0.95$	[0.388, 0.688]	0.300	58.6%	Yes	Yes
Diagnostic AI, $\rho = 0.90$	[0.525, 0.775]	0.250	65.5%	Yes	Yes
Diagnostic AI, $\rho = 0.95$	[0.613, 0.775]	0.162	77.6%	**No**	No

Note. Hidden target = 0.5625. "Eligible" means the complete hidden pairing itself satisfies the score floor, the condition in Proposition 1. Numerical inclusion of the scalar target is weaker. These are sensitivity regions, not confidence intervals. Primary AI removes geography-equality features; diagnostic AI retains them.

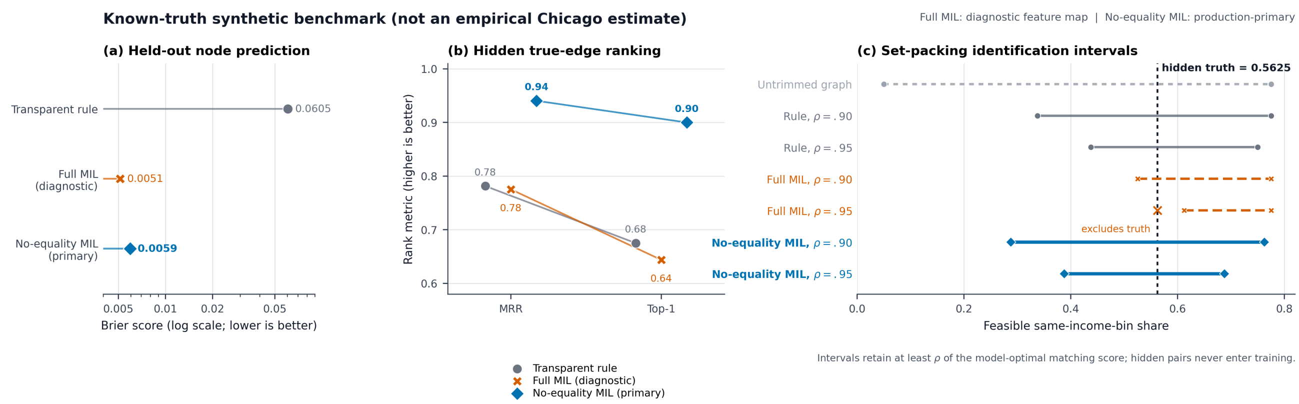

Figure 1: Locked known-truth integration benchmark. Primary AI removes six geography-equality columns; diagnostic AI retains them. Lower Brier is better. The dashed line is hidden truth, never supplied to model fitting. Diagnostic AI looks sharper but its 95% interval misses truth.

Table 6: Candidate/solver check over 20 replicates.

Bin (min)	Edge / truth	Raw width	95% width	Width reduction
1	1.038	0.003	0.003	0.0%
5	1.197	0.020	0.007	66.7%
15	1.563	0.073	0.020	72.7%
30	2.107	0.150	0.027	82.2%

Note. Candidate recall, untrimmed coverage, and 95%-score coverage equal 100% in every row.

Bag-size confounding and calibration. Under noisy-OR, node probability increases mechanically with incident-edge count. A model could therefore predict match status from market density while ranking all incident edges poorly—exactly the distinction suggested by the locked benchmark. We compare against training prevalence, candidate degree alone, and the transparent physical rule; report calibration and Brier loss by degree quintile; and refit with degree-stratified intercepts. Max-pooling and a degree-normalized log-sum-exp aggregator are secondary ablations. Only the predeclared noisy-OR fit determines the main score, and any aggregator selected after viewing Chicago exposure is exploratory.

Optimization and statistical checks. The MILP test suite verifies exact matching, infeasibility, unmatched-node exclusion, SES-gap objectives, score floors, permutation invariance, and agreement with exhaustive enumeration on small graphs. A time-limit incumbent is never labeled an endpoint. Known-truth simulations cross timestamp coarsening, route overlap, SES assortativity, and candidate misspecification; both scalar containment and eligibility of the complete hidden pairing are recorded. Policy uncertainty is recomputed inside date-block resamples instead of bootstrapping a single imputed network.

Table 7: Pre-declared complete-day empirical gates.

Gate	Threshold	Result
Held-out Brier improvement vs. rule	$\geq$ 10%	**TBD**
Candidate-supported matched nodes	report, no target	**TBD**
AI interval-width reduction	$\geq$ 25%	**TBD**
Exact-cover feasibility	$\geq$ 95% analysis cells	**TBD**
Sign stability across candidate grid	same conclusion	**TBD**
Post-policy observation window	$\geq$ 8 months	**TBD**

Predeclared failure outcomes. The paper retains negative and non-identification results. We stop structural estimation for any partial day, surface infeasible cells, report AI failure when the real held-out

Brier gate is missed, and retain the untrimmed range when score restrictions are unstable. If candidate-supported composition changes discontinuously, pre-trends fail, or the policy-effect range crosses zero under reasonable declared rules, the conclusion is respectively measurement shift, descriptive association, or unidentified exposure direction. None is replaced by a maximum-score matching.

7 POLICY ESTIMATION PLAN

Let $Post_t$ indicate trips after January 6, $Zone_i$ the released pickup or path proxy for the newly covered congestion zone, and $Active_t$ the policy's active hours. For outcome Y_{ict} in spatial cell c and date/time cell t, the baseline intention-to-treat specification is

$$Y_{ict} = \alpha_c + \delta_t + \beta(Post_t \times Zone_i \times Active_t) + \Gamma' X_{ict} + \varepsilon_{ict}, \quad (6)$$

augmented by event-time coefficients, zone-specific trends, and weekday/weekend dose contrasts. We first validate treatment timing through discontinuities in `additional_charges`. Outcomes are authorization/all eligible trips and matched/all eligible trips. Inference clusters at spatial and calendar levels where supported by the final cell construction.

The current public table ends June 30, 2026. Production estimation waits until at least September 6 so that eight post-policy months are observable. Before estimating outcomes, we will digitize the official pre- and post-2026 zone polygons, flag trips whose released pickup or drop-off touches the zone, and declare an ambiguity band for spatial units intersected by a boundary. The institutional first stage is the full distribution of `additional_charges` by date, geography, pooled status, day of week, and hour. Specification choices are locked from pre-policy support rather than post-policy outcomes.

Diagnostics include joint and equivalence tests of event-time leads, placebo dates, exclusion of holidays, airports, and special venues, alternative ambiguity bands, tract- and community-area cells, and the weekday/weekend dose contrast implied by the surcharge schedule. We will use a date-block bootstrap or a predeclared small-cluster correction after auditing the number of independent calendar and spatial cells. Failure of pre-trend or overlap diagnostics converts the policy analysis to a descriptive interrupted series rather than a causal result.

For exposure, regressing day-level lower endpoints does not generally produce a lower bound on a policy coefficient. Let $\mathcal{F}_c$ be the feasible pairings in market-time block c and a_c the fixed linear weights induced by the event-study contrast. We propagate latent-edge ambiguity directly:

$$\begin{aligned} \underline{\theta} &= \min_{M_c \in \mathcal{F}_c \ \forall c} \sum_c a_c H(M_c), \\ \overline{\theta} &= \max_{M_c \in \mathcal{F}_c \ \forall c} \sum_c a_c H(M_c). \end{aligned} \quad (7)$$

Date-block resampling will layer sampling uncertainty on this structural range, holding the candidate rule fixed.

The policy is not an instrument that changes only market thickness: it also changes selection into the shared pool. We therefore interpret β as the policy ITT. Thickness and origin composition are mechanisms, evaluated through mediation-style decompositions and sensitivity analysis rather than exclusion restrictions. The event study must pass pre-trend equivalence checks before any causal language is used.

Table 8: Evidence-to-claim ladder for the completed paper.

Claim	Required evidence and boundary
Policy thickened pooling	ITT on authorization/all and matched/all with acceptable pre-trends; not an instrument affecting only thickness.
AI predicts matchability	Out-of-day node calibration beats the rule; does not recover actual co-rider edges.
Candidate exposure range	Certified extrema with support, infeasibility, and threshold audits; conditional on the declared graph.
AI-sharpened range	Score-indexed interval plus known-truth audits and explicit ρ; not a confidence interval.
Policy changed exposure	Structural range has one sign across candidate choices, with block sampling uncertainty; not dispatch causality or realized contact.
Echo chamber	Not testable: no claim about repeated ties, beliefs, or reinforcement.

7.1 Falsifiable scientific hypotheses

The confirmatory analysis separates three questions. **H1 (market thickness)** predicts a positive policy ITT on authorization/all trips and matched/all trips. **H2 (opportunity exposure)** is deliberately two-sided: a thicker pool may lower H by connecting more dissimilar origin neighborhoods, or raise it if added requests form geographically segmented local submarkets. The scientific result is the sign of $[\underline{\theta}, \overline{\theta}]$ in Eq. (7), not the sign from one reconstructed network. **H3 (AI value)** requires the learned score to improve held-out node Brier loss by at least 10% and reduce the primary $\rho = 0.90$ interval width by at least 25% relative to the transparent rule, without reversing the substantive conclusion across the candidate grid. Failing H3 leaves the transparent bound as the primary result; it does not invalidate H1 or the untrimmed version of H2.

The proposed sensitivity grid varies the 30-minute start window to 15 and 45 minutes; pickup radii 4 km to 2 and 6; drop-off radii 7 km to 4 and 10; degree cap 16 to 8 and 32; and $\rho \in \{0, 0.80, 0.90, 0.95\}$. We report one-at-a-time changes and prespecified joint corners rather than choosing a graph after observing exposure. The primary candidate rule is frozen before policy estimation. An infeasible exact cover remains an infeasibility result, not a license to discard inconvenient nodes.

7.2 Mechanism accounting

We report four layers without treating them as a causal mediation chain: (i) authorization changes by pickup-income bin; (ii) matched/all changes by the same bins; (iii) untrimmed physical-compatibility ranges; and (iv) score-restricted ranges. Reweighting post-policy origin, hour, and OD-distance cells to their pre-policy distribution provides a composition-standardized descriptive check. The contrast between untrimmed and score-restricted results measures sensitivity to the compatibility model, not the hidden platform objective. Provider identity, rejected requests, quoted prices, vehicles, and dispatch scores are absent; therefore the public data cannot separate rider selection, supply response, and proprietary assignment decisions.

8 LIMITATIONS AND ETHICAL CONSIDERATIONS

This study uses public, privacy-coarsened administrative records and aggregate census statistics; it recruits no participants and performs no intervention. The applicable institutional determination on human-subjects review remains to be obtained and disclosed before submission. Public availability does not eliminate ethical obligations. First, origins are neighborhood proxies and must not be used to infer a rider's income, race, or identity. Second, candidate edges are relational hypotheses. Publishing trip IDs or a single reconstructed graph could encourage re-identification or stigmatize neighborhoods; released artifacts therefore contain aggregate intervals, synthetic examples, and code, but no row-level inferred social graph. A small-cell suppression threshold will be fixed before release. Third, `trips_pooled` describes a service chain and does not guarantee simultaneous occupancy. Fourth, no stable rider ID exists, so the data cannot measure repeated exposure, belief reinforcement, or an echo chamber. Fifth, model-based score restrictions can under-cover, as the locked 95% result demonstrates. We report untrimmed bounds and a full sensitivity grid alongside any tighter interval.

The City release also suppresses geography non-randomly and rounds time. ACS joins cover neighborhoods, not people, and tract missingness can alter the analyzed subpopulation. We will report missingness by treatment cell, use community-area alternatives, and avoid individual-level fairness claims.

The method can also be misused. A platform could optimize inferred mixing in a way that raises wait, detour, surveillance, or disparate service burdens, or use contextual SES predictions for discriminatory targeting. We present the bounds as an auditing device, not a license to infer sensitive attributes or force interaction. Any policy interpretation must jointly report service quality, geographic coverage, suppression, and distributional burdens. If the effect interval overlaps zero or changes sign across reasonable candidate rules, the scientific conclusion is non-identification.

9 GENERATIVE AI USAGE

Generative AI tools (OpenAI ChatGPT and Codex; exact model identifiers and access dates to be completed at submission) were used to assist code scaffolding and debugging, literature-search organization, prose drafting, and formatting. All research questions, estimands, model restrictions, analyses, source verification, and claims remain the responsibility of the authors. AI-generated code was executed against deterministic tests and known-truth simulations; numerical results in the paper are produced by saved scripts rather than generated text. No confidential participant data or proprietary platform records were supplied to a generative AI system for this draft. Generative AI did not recruit or interact with participants, label empirical observations, or appear as an author. The final disclosure will enumerate every material use across the full research process.

10 CONCLUSION

Privacy protection should not force urban AI research to choose between ignoring relational outcomes and pretending that an imputed network is observed. BoundPoolturns coarsened pooling records into a transparent admissible-set analysis: weak supervision scores compatibility, set packing enforces structural consistency, and optimization propagates ambiguity to an opportunity-exposure estimand. The locked benchmark shows both the promise and the danger of this approach. A proxy-audited AI score tightens the interval and retains truth in the locked test, whereas a shortcut-rich score looks sharper and confidently excludes truth. The next empirical step is therefore not another synthetic benchmark; it is the pre-declared complete-day Chicago analysis in Table 7. Until those gates are met, the method is a validated pilot rather than evidence that ride pooling changes urban social mixing.

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A REPRODUCIBILITY APPENDIX

The artifact contains the complete candidate builder, weak noisy-OR model, MILP bound solver, locked synthetic generator, design lock, unit tests, and machine-readable outputs. The current implementation uses NumPy, pandas, SciPy/HiGHS, scikit-learn, and Matplotlib. Seven set-packing tests cover exact matching, infeasibility, score restriction, SES gaps, unmatched-node exclusion, synthetic candidate recall, and agreement between the MILP and an exhaustive small-graph fallback.

The complete-day extraction requests every authorized row for two comparable weekdays, checks server-side counts, writes atomically, hashes files, and audits duplicate IDs, window boundaries, match/pooling consistency, and spatial missingness. The current execution environment blocked both supported City API routes; no row count or partial complete-day file is reported in this draft.

B PROOF OF PROPOSITION 1

The feasible vectors in Eq. (4) and exact-cover pairings on the declared graph are in one-to-one correspondence. Because this set is finite, certified global minimization and maximization attain sharp endpoints. If $z^\dagger$ is feasible, its value lies between them; after adding the score floor this holds exactly when $\sum_e w_e z_e^\dagger \geq \rho W^*$. Finally, $\rho_2 \geq \rho_1$ implies set inclusion, so lower endpoints weakly rise and upper endpoints weakly fall, while membership of $z^\dagger$ need not persist.

C IMPLEMENTATION PSEUDOCODE

C.1 Degree-capped candidate graph

(1) Split by date and train/test assignment; project released centroids.
(2) Build a normalized start-time/pickup-coordinate k-d tree.
(3) Query at most 96 neighbors, apply hard filters, and retain 16 proposals.
(4) Deduplicate by physical cost while enforcing degree at most 16.
(5) Emit support/degree diagnostics; forbid cross-date or cross-split edges.

C.2 Out-of-day weak edge scoring

(1) Construct compatibility features without labels, fares, or SES.
(2) Fit the rule intercept/scale on supported training nodes.
(3) Fit $\widehat{\theta}$ using the same node likelihood and no edge label.
(4) Compute held-out node probabilities and calibration metrics.
(5) Compute reciprocal rank and top-k only in known-truth simulations.

C.3 Certified exposure range

(1) Verify graph integrity, finite scores, parity, and feasibility.
(2) Solve $W^* = \max_{z \in \mathcal{M}(E_C)} \sum_e w_e z_e$ to global optimality; add the score floor only for a declared ρ analysis.
(3) Solve both SES extrema; reject uncertified time-limit incumbents.
(4) Normalize by pair count and retain certificates/configuration.